PREFACE

Name	Extracorporeal Shockwave Lithotripter
Model	HK.ESWL-V
Documentation	Operating Manual
Revision	Ver1.0
Date	2010.05
Language	English

Operating manual

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List of revisions

Date	Revision	Content
05/2010	Original	

Declaration of conformity

Manufacturer	Shenzhen Huikang Medical Apparatus Co., Ltd
Address	3/F, Shenyao Building, Bagua 3rd Road, Shenzhen 518029 P.R. China
Product	Extracorporeal Shockwave Lithotripter
Model	HK.ESWL-V

Classification

I B Equipment (GB9706.1-1995« The People's Republic of China National Standard for Medical Electrical Equipment Part one General Requirements for Safety)
a) Type I device for the anti-shock type;
b) Type B device for the degree of protection against electric shock;
c) Belonging to ordinary type;
d) Continuous operation of the equipment;
e) This equipment is not used in a flammable anesthetic mixture of gas, air, oxygen or nitrous oxide gas or gas mixture;
f) The device has a signal input part;
g) The device does not have defibrillators, discharge parts.

Standard

Applicable standards as below:
GB191-2000, GB6920-86, GB8279-87, GB9706.1-1995, GB9706.3-92, GB9706.3-2000, GB9706.11-1997, GB9706.12-1997, GB9706.14-1997, GB9706.15-1999, GB9706.22-2003, GB9969.1-1998, GB10152-1997, GB/T13384-92, GB/T14710-93, YY0001-90

Reading illustration

WARNING **CAUTION** （mentioned in following chapters）

CAUTION	Do always follow the instructions to prevent injury to human body or damage to the lithotripter. Violation to the Caution may lead to slight injury to human body.
WARNING	Do always follow the instructions to prevent injury to human body or damage to the lithotripter. Violation to the WARNING may lead to serious injury to human body.
Authorized personnel	The staff trained by Shenzhen Huikang Medical Apparatus Co., Ltd or by the company which is authorized by HK.

Operating manual must be read and comprehended by the user before operating the HK.ESWL-V.

All the users of Extracorporeal Shockwave Lithotripter HK.ESWL-V must conform to all the rules of operating manual.

Declaration for localization system configuration

Localization system configuration

There are two kinds of localization system for the shockwave lithotripsy machine. They are X ray localization system and Ultrasound localization system respectively.

1.X ray localization system (Standard)

The registered main unit of HV generator, X ray tube assembly, image intensifier, HV cables used as combination of X ray

2. Ultrasound localization system (Optional)

Ultrasound localization system is optional, it is just a mechanical assembly which is used to make ultrasound probe move around the shockwave source and measure the distance from the focus to the surface of probe. Ultrasound scanner is not included.

As outlined ultrasound, users can choose ultrasound scanner machines with registration certificates according to local requirements.

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FOREWORD

Introduction

The foreword consists of the following contents:

- Equipment introduction
- Main functions and application range

Equipment introduction

HK.ESWL-V is a set of shockwave device mainly used for fragmentation of urinary calculi.

HK.ESWL-V includes Electromagnetic/ Electro hydraulic shockwave source, C arm X ray localization system and ultrasound localization (optional)

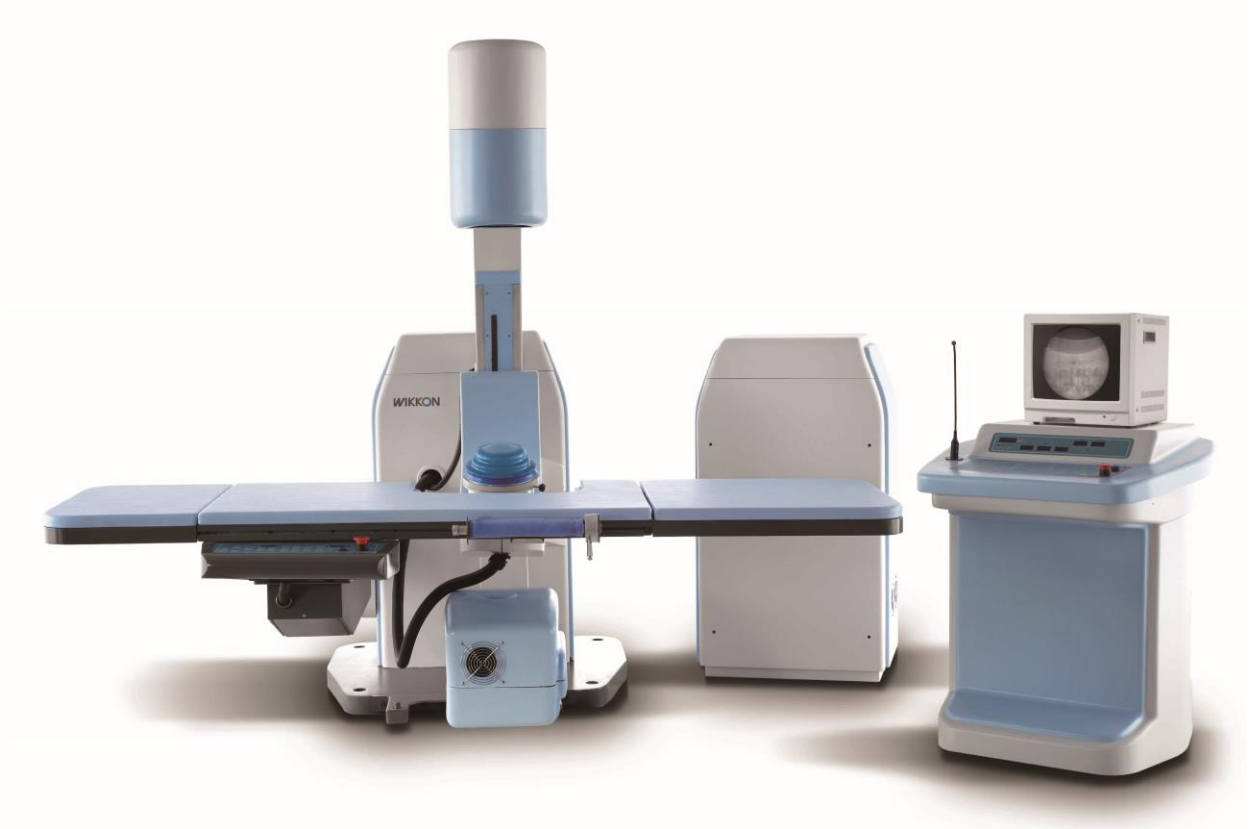


Figure 1 HK.ESWL-V

Medical application

Medical application for HK.ESWL-V as follows

Application range

Stones in urinary system

Contraindications of shockwave application

Before shockwave therapy, attending doctors must examine if patient have the following contraindications, then make the right decision according to lithotripsy treatment risk

Contraindications including:

- Shockwave through cardiac muscle tissue and central nerve
- Shockwave through lung tissue
- Shockwave through intestinal gas
- Shockwave through tumor
- Shockwave through Aneurysm
- Other undefined symptoms

Relative contraindications includes:

- Pregnancy
- Hemorrhagic disease Oliguric renal failure
- Decompensation of the cardiovascular disease
- Uncontrollable urinary track infection
- Patients with upper urinary tract anatomy, abnormal shape affecting stone-discharge, such as ureteropelvic junction obstruction, calyceal neck stenosis, distal ureteral stones narrow

Additional contraindications including:

- Polyneuropathy

It is the responsibility of the attending doctors to define indications and adopt applicable treatment procedures according to the latest modern medical knowledge. Before the treatment of a patient implanted pacemaker, the possibility of shockwave therapy should be assessed according to the published criteria so far. Shockwave therapy can be accepted only in case of no risk.

Technical information

Shockwave Generation	Electromagnetic: Using a plane coil loaded with high-voltage pulse and a piece of metal membrane to generate shock waves, the structure of shock waves are called electromagnetic shock waves. Electro-hydraulic: Shock wave is generated through discharging of electrode under water, the structure of shock wave are called electrohydraulic shock wave.
Focusing	Electromagnetic: Shockwave generator gives shockwave which is vertical to electromagnetic disk, through conductivity of water, and using an acoustic lens to focus the shock wave. Electro-hydraulic: Using the pressure of pulse loaded into both ends of electrodes, then generating discharging to ground, further generating shock wave, and using reflector to focus the shock wave.

SECURITY GUIDE

Introduction

The safety section describes the safety regulations that must be abide by when operating HK.ESWL-V.
The contents as follows:

- Responsibility of Shenzhen Huikang Medical Apparatus Co., Ltd
- Responsibility of the users
- Electrical installation of HK.ESWL-V
- Safety for patients
- Use of Electromagnetic/Electro hydraulic shockwave
- Precautions before shockwave release
- Hazardous area
- Safety for operating staff
- Safety of + HK.ESWL-V

Responsibility of Shenzhen Huikang Medical Apparatus Co., Ltd

Regardless of the specific requirements set by the relevant liability standards, Shenzhen Huikang Medical Apparatus Co., Ltd assumes responsibility under the following conditions only:

Assembly, adjustment, maintenance and modification are performed only by personnel authorized by Shenzhen Huikang Medical Apparatus Co., Ltd.

Electrical installation of treatment room complies with national standards locally.

Operation of HK.ESWL-V must be in accordance with requirements of valid operating manual.

Responsibility of manufacturer

Regardless of the specific requirements set by the relevant liability standards, Shenzhen Huikang Medical Apparatus Co., Ltd assumes responsibility under the following conditions only:

Assembly, adjustment, maintenance and modification are performed only by personnel authorized by Shenzhen Huikang Medical Apparatus Co., Ltd.

Responsibility of users

Before starting HK.ESWL-V, users must read and understand the user instructions.

All the users who operate HK.ESWL-V must conform to all the provisions of technical and operating manual.

To operate HK.ESWL-V only in an appropriate area where the electrical installation must comply with applicable national standards.

The installation room shall comply with applicable local regulations of X ray radiation prevention.

Safety for patients

To avoid patients or others at risk, users have to make sure that HK.ESWL-V is in good working conditions before operating it. It is responsibility of operating staff to tell patients about the treatment process in advance. In particular, patients must be told prior to lithotripsy treatment about the noise caused by release of shockwave. This will prevent patients from being startled and making involuntary movement during treatment.

Safety for operating staff

- Training operating staff on HK.ESWL-V

Shenzhen Huikang Medical Apparatus Co., Ltd offers comprehensive training on HK.ESWL-V for physicians and operating staff.

Under no circumstance HK.ESWL-V could not be operated by untrained or unqualified personnel. Training is provided by authorized staff of Shenzhen Huikang Medical Apparatus Co., Ltd.

- Duty of operating staff

Operators and cleaners should be very careful when operating and cleaning HK.ESWL-V.

Before activating HK.ESWL-V, operators must completely read and understand the operating manual.

Do not operate HK.ESWL-V in an explosive or inflammable atmosphere, which may be caused by volatile gas from anesthesia, detergents, or disinfectants.

- Cleaning and disinfecting of HK.ESWL-V

Disconnecting power supply before cleaning and disinfecting HK.ESWL-V. With non-corrosive latex disinfectant or disposable surgical towel mulching, replacement for each treatment with purpose of avoiding cross infection caused by contact of hepatitis B bacteria and skin disease.

Safety of HK.ESWL-V

Electromagnetic compatibility

HK.ESWL-V will comply with electromagnetic compatibility requirements of IEC601-1-2 without release of shockwave. According to the standards of IEC60601-2-36, it will be acceptable that EMC of HK.ESWL-V may not comply with IEC601-1-2 during shockwave triggering.

During shockwave triggering, keep HK.ESWL-V away from ECG monitor under working condition.

Atmosphere environment

Do not operate HK.ESWL-V in an explosive or inflammable atmosphere, which may be caused by volatile gas from anesthesia, detergents, or disinfectants.

Equipment electrical installation

Electrical safety conditions for installation must comply with applicable national standards.

The circuit breaker conforms to the requirement of GB9706 will be installed between equipment and grid, by which disconnection between equipment and grid will be realized. Power supply of HK.ESWL-V is equipped with circuit breaker which can cut off phase wire. The circuit breaker should conform to the standards of IEC328, moreover, it is non-self resetting. The capacity of fuse required should not be less than 10A.

HK.ESWL-V will not be allowed for parallel application in the same wire with other large load equipment instantaneously (welding machine, elevators etc).

HK.ESWL-V requires the installation of functional earthing and protective earthing, equipped with separate ground wire. Ground resistance ≤4 Ohms.

WARNING

HK.ESWL-V shall be installed by authorized personnel of Shenzhen Huikang Medical Apparatus Co., Ltd.

Protective earth terminal and functional earth terminal shall be connected properly before HK.ESWL-V starts.

Console operation in a separate room; construction for the installation of main unit must comply with the requirements of X ray radiation prevention and be approved by test of local government department.

After installation or re-installation, electrical safety test must be

performed for HK.ESWL-V.

For details, please refer to chapter one "Safety Inspection" of operating manual.

Transportation of HK.ESWL-V

Before transportation of HK.ESWL-V, outer connection with cables and water pipes must be cut off.

Do not move HK.ESWL-V through slopes with more than 15 degrees or stairs with height difference of more than 2mm.

Use of HK.ESWL-V

CAUTION

Shockwave triggering may cause interference to other device.

HK.ESWL-V will not be allowed for parallel application in the same wire with other large loads equipment instantaneously (welding machine, elevators etc).

During shockwave triggering, keep HK.ESWL-V away from other electronic device like ECG monitor.

WARNING

Shockwave should be triggered in coupling state only.
Reflection and refraction at contact surface between coupling cushion and air may damage coupling cushion, furthermore, it will affect the lifetime of cushions.

Power description

HK.ESWL-V requires power of single phase, 220V, 50Hz. Power supply of HK.ESWL-V is equipped with the circuit breaker that can cut off phase wire, the circuit breaker must conform to the requirements of IEC328, and it is non-self resetting. Fuse capacity of the circuit breaker should not be less than 15A.

Allowed fluctuation range for power voltage: 220V±22 V
Allowed fluctuation range for power frequency: 50Hz±1 Hz

HK.ESWL-V will not be allowed for parallel application in the same wire with other large load equipments instantaneously.

WARNING HK.ESWL-V will not be allowed for parallel application in the same wire with other large load equipments instantaneously (welding machine, elevators etc).

Electrical installation of HK.ESWL-V

a) Circumstance temperature 15～35℃
b) Relative humidity 45%～85%
c) Atmosphere pressure 86～106KPa
d) Power requirements AC 220V±10%, 50±1Hz, internal resistance no more than 0.3Ω
e) Fuse 15A
f) Space requirements:
- Main unit room ≥ 25m², Height > 2.5m
- Console room ≥12m² (Separate room)
- Main room for installation of HK.ESWL-V requires X ray radiation prevention (Please refer to GB9706.12 requirements for detailed standards; inspection by local quarantine department)

WARNING Protective ground line should be connected well to ensure machine safety. HK.ESWL-V uses three-phase four-wire power supply. Original phase must be restored if supply grid need to be maintained, or it will cause the

failure of HK.ESWL-V.

WARNING	**HK.ESWL-V must be installed by authorized personnel from Shenzhen Huikang Medical Apparatus Co., Ltd.** **Protective ground and functional ground must be connected well before starting HK.ESWL-V.**
WARNING	**After installation or re-installation, the electrical safety for HK.ESWL-V must be tested.** **Please refer to the chapter of "Safety Inspection" of operating manual for details.**

Safety for patients

- Operation of HK.ESWL-V

  Do not operate HK.ESWL-V in an explosive or inflammable atmosphere, which may be caused by volatile gas from anesthesia, detergents, or disinfectants.

- Informed consent of patients

  Patients must be informed about the treatment process. In particular, patients must be informed about the noise caused by the release of shockwave. This will prevent patients being startled and making involuntary movements during treatment.

Use of electromagnetic shockwave

Physicians who use shockwave in therapy must be familiar with the medical aspects of the process, including currently valid indications and contraindications as well as corresponding adverse effects. Especially, but not exclusively, patients with coagulopathy must be excluded from shockwave treatment.

If shockwave are directed into air-filled organs, such as the intestinal tract or the lungs, tissue damage can be caused. Particular caution is recommended in order to avoid shockwave into lungs.

Energy of shockwave generated by the lithotripter through muscle tissue will be attenuated, and its remaining energy will be absorbed by bones.

There is direct relationship between shockwave dose (shockwave intensity times shocks number) and the appearance of hematomas. But it is not enough to draw conclusion that higher triggering frequency of shockwave will influence the occurrence of side effects. Although this has not yet been definitely proven by scientific means, medication, age, blood pressure or vascular deficiencies, especially in the renal area, can increase the risk of side effects during exposure to shockwave. The sensitive vessels in the renal parenchyma are particularly at risk, since lesions caused by excessive shockwave intensity can result in hematomas. For this reason, the shockwave dose must always be maintained as low as possible. As the shockwave intensity increases, the release frequency must be decreased.

Before releasing shockwave, ensure good contact between patients' body and coupling cushion in the shockwave impact area. If air bubbles exist, the efficiency of shockwave will be attenuated, which will lead to unsatisfactory treatment effect even skin erythema.

Lithotripsy treatment for patients with other inflammation or diseases may be influenced, even worse after lithotripsy. It is necessary to cure these diseases before lithotripsy.

WARNING

Heart under pressure impulse will cause harmful activities!
Patients should be examined regularly by attending physicians during treatment.

WARNING

Any hazardous movements from patients or operating staff should be prohibited during lithotripsy treatment.

Precaution of shockwave triggering

Shockwave release will generate noises. It is recommended that operating staff and patients all wear earplugs during lithotripsy treatment.

WARNING

Pressure impulse will bring harmful heart action.

Shockwave through air-filled organs such as intestinal, lung etc. will bring tissue damage.

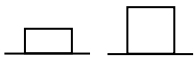
Lung and heart impact from shockwave must be avoided.

Operating staff must note possible movements carefully, which may hurt operating staff and patients.

Device labels

Table 1 shows the labels on the HK.ESWL-V.

Table 1 WARNING and information labels

Labels	Meaning
	Observe operating manual
	Risk of contusion!
	Type B equipment
	Nameplate Device's and manufacturer's reference includes: Name of manufacturer: Shenzhen Huikang Medical Apparatus Co.,Ltd. Name of device: Extracorporeal Shockwave Lithotripter Model of device: HK.ESWL-V SDA Registration No. Input power: 5kVa Input voltage: AC220V，50Hz Address of manufacturer: 317, Shenyao Building, Bagua 3rd Road, Shenzhen
	Functional earth terminal Need to be connected well while installation
	Protective earth terminal Need to be connected well while installation
	Power button indicating marks: push the button down to the lower position to power on the system, and push a second time to release the button to a higher position to power off the system.
~220V	(Appears on the nameplate) Refers to alternating current electricity power.
	Ionizing radiation label

DEVICE DESCRIPTION

Introduction

The description section contains the overview and description of functional parts and controls of HK.ESWL-V.

Parts and view of HK.ESWL-V

Figure 2 shows the HK.ESWL-V and its functional parts.

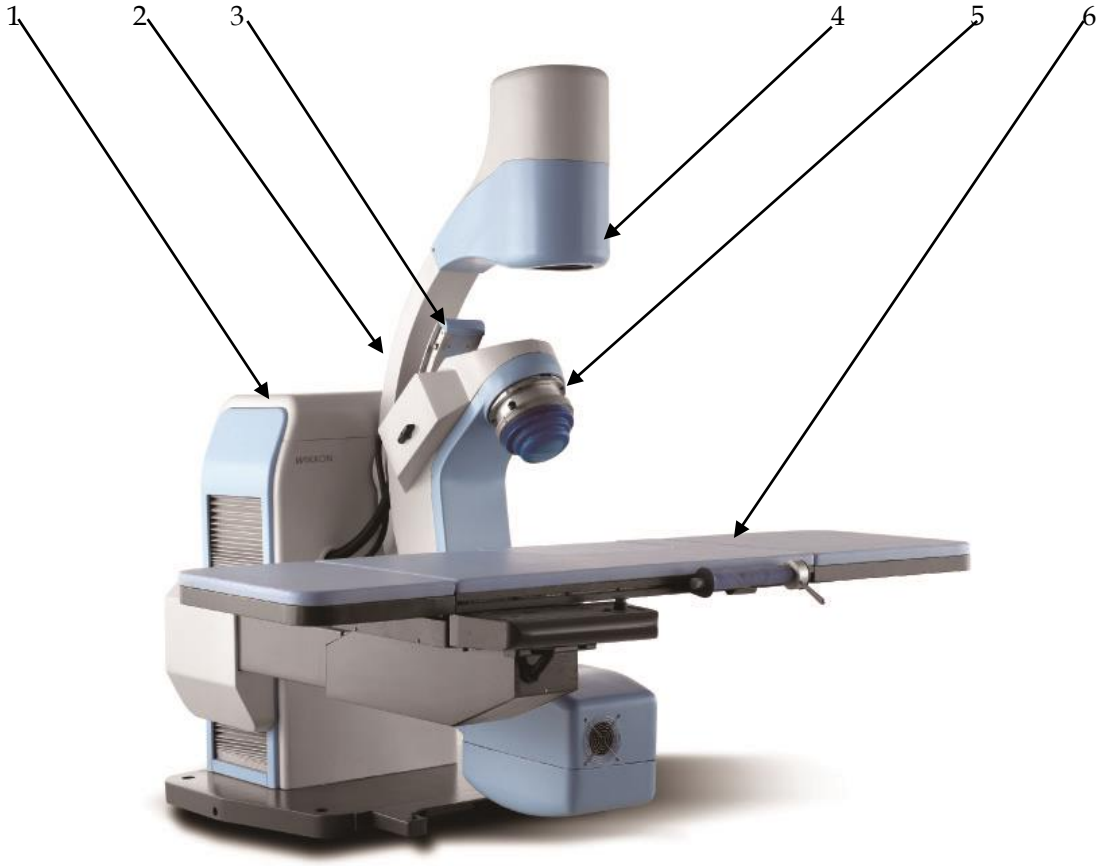


Figure 2 HK.ESWL-V

1. Main unit
2. C arm
3. Membrane arm
4. Image intensifier
5. Shockwave source
6. Table

Parts function description

Parts	Function	Relations with other parts
Main unit	Supporting matrix for C arm and membrane arm motion, shockwave source	Assembling C arm and membrane arm, shockwave source, image intensifier, X ray tube, ultrasound localization device
Table	Achieving diagnosis of patient, treatment body position as well as motion of table	Assembling belly band
Table-side hand piece	Control parts of main unit motion, shockwave generator, ultrasound localization	Connecting with main unit, table, motion arm, shockwave generator and ultrasound localization system through cables, achieving lithotripsy function.
Console	Control part for motion, shockwave generator, X-ray localization	Connecting with main unit, table, motion arm, shockwave generator and X-ray localization system through cables, achieving lithotripsy function.
Shockwave source	Core parts of lithotripter for generating shockwave and breaking urinary stones	Installed in lower end of membrane arm, Shockwave source cup and ultrasound localization for mechanical connection
Membrane arm	Supporting parts for shockwave source motion, driving shockwave source rotating, facilitating treatment of urinary stones in different locations.	Integrated with main unit shockwave source installed in lower end
C arm	Supporting parts for X ray image system; rotating, driving X ray image system(image intensifier, X ray tube) do	Integrated with main unit, X ray tube and image intensifier installed in upper and lower ends respectively.

	spherical movement around focal point of shockwave source.	
Image intensifier	X ray images receiving and processing parts	Installed in the upper end of C arm; synchronized with X ray tube to move spherically around focus point of shockwave source.
X ray tube	X ray generating parts X ray localization assembly parts	Installed in the lower end of C arm; in the upper end of tube installed calibrator. Synchronized with image intensifier to do spherical movement around focus point of shockwave source.
Calibrator	To control direction of X ray beam	Installed in the upper end of tube
Ultrasound localization assembly	(optional） Mechanical device, driving ultrasound probe to do the straight-line distance-measuring around focus point of shockwave source. Ultrasound scanner machine and probe are not included	with shockwave source cup for mechanical connection
Monitor and PC	Observing, holding and storing X-ray images etc.	Putting on the console, connecting with image intensifier signal
Belly band (accessory)	(accessory, optional) Fixed for patients receiving treatment	Installed in the table

Patient Table

Patient table of HK.ESWL-V as shown in the figure 4
Loading capacity: 135kg

Figure 3

1. Table assembly
2. Belly band
3. Table-side operating panel

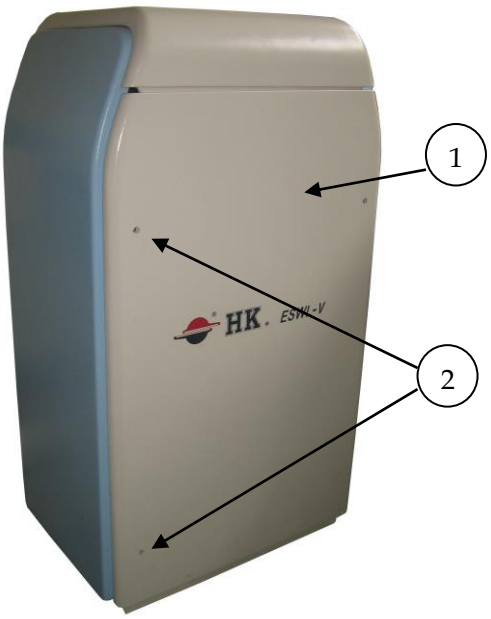
Electrical control cabinet

Figure 4 shows electrical cabinet of HK.ESWL-V.

Cabinet door may be opened to facilitate maintenance of HK.ESWL-V if needed.

There are screw holes in the cabinet door, and there is bolt tightened to prevent the cabinet door from opening wrongly.

Figure 4



1. Electrical cabinet door
2. Screw hole

WARNING

Only authorized persons can open the electrical cabinet door.
Authorized persons mean:
Personnel trained by Shenzhen Huikang Medical Apparatus Co., Ltd; or by companies authorized by Huikang.

Turn off the power before electrical cabinet door opening.

cut

WARNING

Before maintenance, the installation of functional grounding must be compulsory and permanent.

Make sure high voltage capacitor discharge completely before functional grounding wire disconnected.

Remote Console

Figure 5 shows console of HK.ESWL-V.
Electrical console can be opened for maintenance if needed.

Figure 5

Shockwave source

Electromagnetic shockwave source

1. Cushion 1
2. Cushion hoop
3. De-gassing tube
4. Ultrasound turntable
5. Top screw for ultrasound turntable

Hazardous area

For the safety of operators, the HK.ESWL-V is designed according to strict safety criteria. Nevertheless, please pay special attention to dangerous areas as shown in figure 6. When the table of HK.ESWL-V extracorporeal shockwave lithotripter is in motion, close to danger areas may lead to bump and crush.

Figure 6 Hazardous area

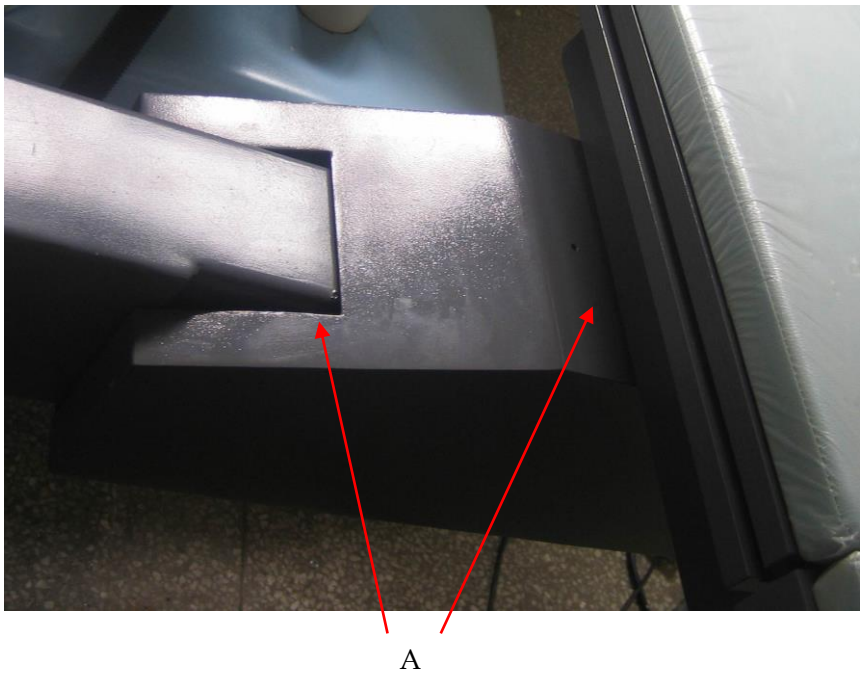


Table 2 Hazardous area

Description	Reason	How to avoid
A: Crush danger to operators and patients	The hands are too close to A when HK.ESWL-V is operated.	Keep away from area "A"

Operating panel

The operating panel offers all function buttons of HK.ESWL-V.
Figure 7 and Figure 8 show all the buttons and indicators of the operating panel.

Figure 7 Operating panel for console

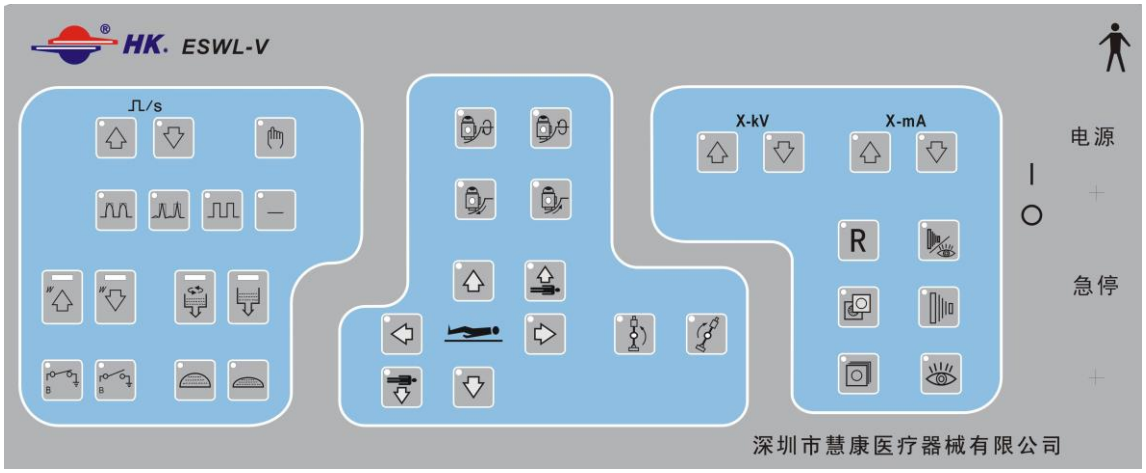


Figure 8 Display panel for console

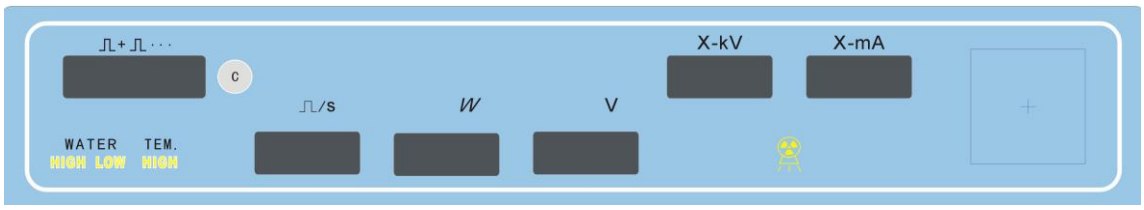
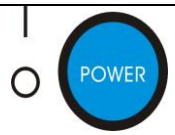
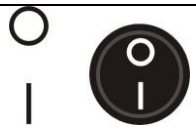
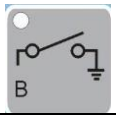
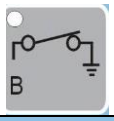
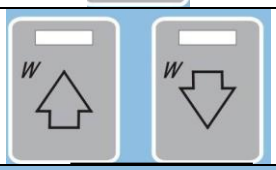
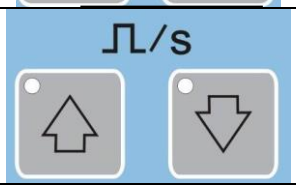
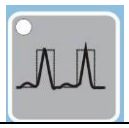
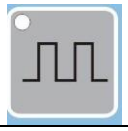
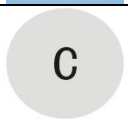


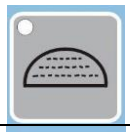
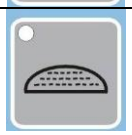
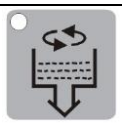
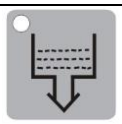
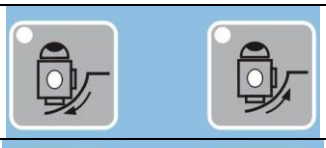
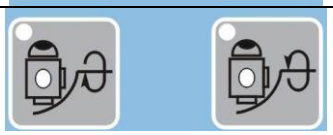
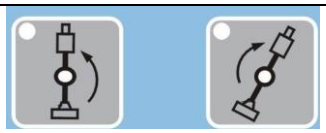
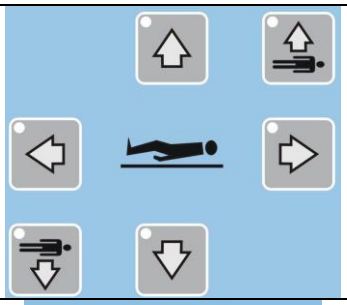
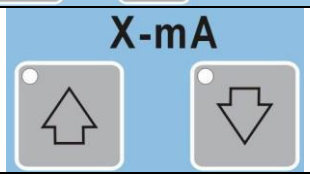
Figure 9

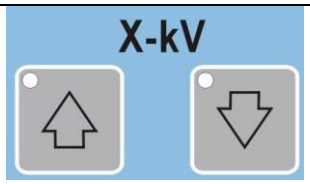
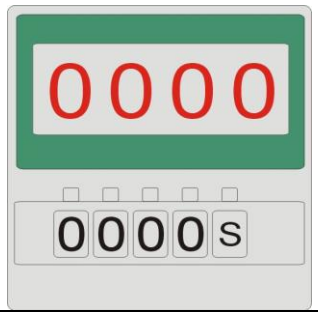
Table-side operating panel

Table 3 Buttons and indicators function

Icons	Name and function description

	**Power switch** **Press the button to "1" position to switch on the lithotripter, and press again to "O" position to switch off the lithotripter**
	**Power switch** **Press this button beside table to "1" position to turn on control power beside table, and press again to "O" position to switch off.**
	Emergency switch Press this button to turn off power, clockwise rotation to turn on the power
	**High voltage start button** **Press down to connect high voltage of shockwave**
	**High voltage stop button** Press this button to disconnect high voltage, and self-reduction voltage, grounding, capacitor discharging
	**Energy up/down button** Press the buttons to increase/decrease energy of shockwave triggering, it indicates shockwave intensity
	**Shockwave triggering frequency button** Press the buttons to increase/decrease frequency of shockwave triggering.
	**ECG triggering** Connecting to ECG synchronization signal, press the button to generate synchronization shock wave
	**Continuous triggering** Press down the button to generate continuous & intermittent shockwave
	Manual triggering button Press down the button for single shockwave triggering
	**Reset button for counting** Press this button to reset counting to zero

	**Water in button** Press and hold the button to add water into the cushion
	**Water out button** Press and hold the button to drain water from the cushion
	**Water cycling start button** Press this button for continuous water cycling and air-out
	**Water cycling stop button** Press this button to stop water cycling and air-removing
	Probe up button Press this button to move probe up, motion distance 100mm
	Probe down button Press this button to move probe down, motion distance 100mm
	**Up/down movement of membrane arm** Press the button for up/down orbital movement of membrane arm, range 0-30 degrees
	**Membrane arm rotation around axis** Press the button for clockwise and counterclockwise movement of membrane arm, range 0-180 degrees
	**C arm clockwise movement button** Press the button for clockwise and counter clockwise movement of C arm, range ±30 degrees
	**Three dimensional movement of table button** Press the button to start three dimensional movement of table: Up/down motion range 120mm, accuracy ±1mm Right/left motion range 100mm, accuracy ±1mm Backward/forward motion range 100mm, accuracy ±1mm
	**X-mA adjustment button** Press the buttons to increase/decrease fluoroscopy mA value. Step value 0.1mA, adjustment range 0-5.0mA

	**X-kV adjustment button** Press the buttons to increase/decrease fluoroscopy kV value. Step value 0.1kV, adjustment range 0-5.0kV.
	Exposure button Press and hold the button for fluoroscopy
	Energy indicator To indicate shockwave triggering voltage, that is shockwave intensity. Unit is kV. Shockwave range is level 0-16, step value 1 energy level.
	Shockwave triggering counting display Showing shockwave triggering times Counting range 0-9999 times, step value one time
	Triggering frequency display Showing shockwave triggering frequency, frequency range is 60-120 times/minute
	Distance-measuring display for ultrasound probe Showing the distance between probe and focus of shock wave, distance-measuring range 25-125mm
	Power voltage display Showing network power voltage AC230V
	X-KV voltage display X-KV voltage pre-set display, voltage range 50-100KV
	X-mA current display X-mA current display, current range 0-5.0mA
	**X ray exposure limit timer** X ray exposure limit timer preset value is 0-300 seconds. Exposure duration for single time less than 300 seconds

OPERATION

Introduction

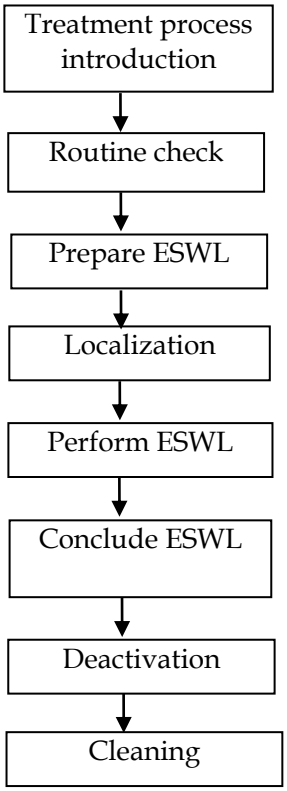
The operation section informs you about:

- Which safety instructions must be observed
- How to activate HK.ESWL-V
- Which rules must be observed before starting HK.ESWL-V
- How to use HK.ESWL-V for treatment
- How to check localization accuracy
- How to make HK.ESWL-V in state of dormancy
- How to clean HK.ESWL-V

Safe instructions described in the safety section must be observed.

Figure 13 shows the operating procedure of HK.ESWL-V.

Figure 13 Operating procedure



Observing safety instruction and regulations

Before operating HK.ESWL-V, it is necessary to read and observe the safety instructions and regulations given in the operating manual.

Observing regulations in the safety sections
Observing relevant national regulations concerning operation of shockwave devices

Routine check

Routine check should be carried out before activating HK.ESWL-V, which helps to avoid risks caused by device damage.

Visual check

Check the following parts to see if there is any signs of damage:

- Cushion
- C arm and table
- Outer cables and connections
- Control panel

WARNING

Do not activate HK.ESWL-V if the protective earth terminal is not properly connected.
Only authorized personnel could conduct the above-mentioned visual check. Authorized personnel are exclusively persons who have been trained by Shenzhen Huikang Medical Apparatus Co., Ltd or by a company authorized by Huikang.

WARNING

Do not activate HK.ESWL-V if any defect or damage is found during routine check.
Perform precaution check before ESWL treatment, then trial run, lastly perform the ESWL treatment after ensuring safety working.

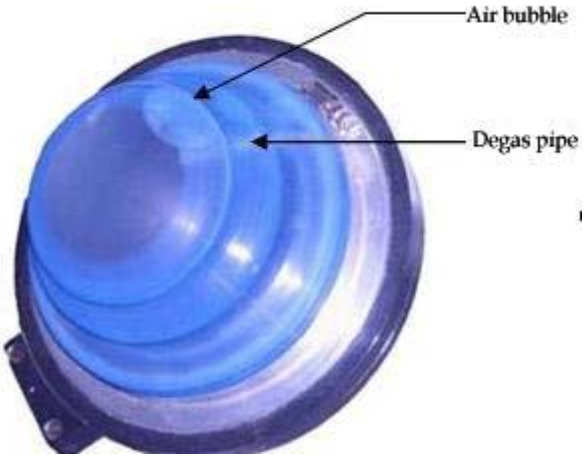
Degas of the cushion

If there are air bubbles inside the cushion, please remove them as per following instructions:

Option 1: Starting water cycling for self-degas

Option 2:

1) Adjusting the therapy head to such a position that de-gas pipe of cushion is on the top.
2) Untying de-gas pipe of cushion and hold it upward
3) Tapping on the cushion to de-gas
4) Re-tying de-gas pipe after all the air bubbles are removed.



WARNING

Do not release shockwave if there are air bubbles inside the cushion. At the same time, operators have to use coupling gel properly to avoid air bubbles between cushion surface and human skin.

Preparation before ESWL treatment

Patient's preparation

Patients who are diagnosed with urinary stones and suitable for ESWL treatment should make the following preparation in advance:

1) Routine check
   - Abdominal plain film(for details of stones)
   - Ultrasound examination
   - Intraveneous pyelography(IVP confirmed renal function)
   - Routine blood test
   - routing urine test
   - ECG
2) Others diseases check
3) Mental preparation: detailed explanation to eliminate patients' fear of the treatment.

WARNING

Patients with heart disease and hypertension should wait for ESWL treatment until they are in stable conditions. Use ECG R wave synchronization trigger if needed. (only use ECG monitor designated by the company to meet the requirements GB9706.1; signal input/output interface must be synchronization connection).

Patients suffering from urinary tract inflammation should have anti-inflammation before the ESWL treatment.

Lithotripsy treatment may be affected if the patient is also suffering from other diseases. The patients could not have lithotripsy treatment until he/she is cured against these diseases.

WARNING

Water temperature should be set to such degree that makes patients feel comfortable with the cushion. The lower water temperature in the cushion is caused by lower environmental temperature (WARNING: temperature inside the cushion can never exceed 40ºC). Heating will be stopped automatically when the temperature reach the setting value, at this time, add water into cushion once again. Heating temperature is set to be 30ºC during installation.

Preparation for shockwave treatment

1. Disinfection of cushion surface
2. Power on the system
3. Press the "Start" button and connect to high voltage
4. Press the "Energy-up" button to raise energy level of shockwave to an appropriate value[1].
5. Press "Single release[1] of shockwave" button several times to run the system on trial. Make sure everything is in order.

The below principles should be followed during implementation of lithotripsy:

1. The stone nearest to exit should be first broken when patients have multiple stones.
2. The stone inside of poor renal function should be broken when there are stones in double urinary track.
3. For single large stones, break the portion nearest to exit first.
4. It is usually within 3000 times discharge for each lithotripsy treatment. Discharge times could be increased properly on the basis of actual lithotripsy need if patients are in good condition.
5. For the same patient, the time interval should be 7 days above for two sessions of ESWL treatment.

[1] Appropriate value means comfortable treatment effect. Because different patient has different sensitivity, it is not a fixed value. Usually this value ranges from 12 to 14 for Electromagnetic ESWL and from 5 to 7 for Electro-hydraulic ESWL.

Localization

Accurate localization is the fundamental guarantee for lithotripsy treatment. HK.ESWL-V is assembled with X ray localization system (Ultrasound localization system optional).

X ray localization

Localization methods of X ray localization system:
Before X ray localization, location, number, shape of stones shall be confirmed by means of X ray plain abdominal radiograph or other examination methods, as well as epidermis stone. After that, make sure that everything is in good condition; The patient can be allowed to lie on the table for next step operation, to apply ultrasound coupling gel on the cushion.

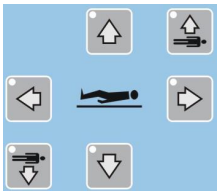
1) Preparation before localization

①Press button, move X-ray C arm in the place of orthographic projection

②Move stone position of the patient in the projection surface of image intensifier.

③Open the X ray television imaging system, press fluoroscopy button , adjust X—kV, X-mA based on clear X ray image for fluoroscopy. (Note: Do not hold this button for a long time.)

1) Localization operation

① To search for stones, press 
, , , buttons， move the patient acrossa flat surface, make stone place in the focus mark of X ray image. press button，move C arm to the side, press button, if stone shown in the image deviate from focus mark, press or button till stone overlaps focus mark again.

CAUTION

press button to fill water into the cushion. Please note that cushion should not be filled too much so as not to raise body position to make the stone deviate from focus mark. Repeat step①.② for calibration, confirm whether the stone deviate from focus mark.

After completion of above-mentioned steps, X ray localization comes to an end. Now starts triggering treatment.

CAUTION

Output center of X ray tube for this device is always fixed in the corresponding input center of image intensifier permanently. Deviation for benchmark axis does not exist. The C arm can be used for fluoroscopy examination at different angle.

CAUTION

If X ray system stops working for more than 3 months, please preheat to make it back to normal working parameters before use. Preheating operation as follows: set 65kV, 3.0mA for exposure 5 minutes, then rest 5 minutes, in this way repeat 5 times.

WARNING

mA as low as possible and reasonable to reduce dose absorbtion of patientsmA

（Options）

Ultrasound localization system

Ultrasound localization system optional, it is only a mechanical assembly which drive probe to move around the focus of shockwave source for straight line measurement. Ultrasound scanner machine and probe are not included in the standard package.

As ultrasound machine of separate use, it can be chosen freely by the users with registration certificate.

Localization method in ultrasound localization system:

CAUTION

Before the patient lies on the table, the operator shall use the ultrasound probe to check the calculus's location, number, and shape. If necessary, mark its best detecting position on the patient body. Then set a proper location of the shockwave head, the probe and the cushion according to the detecting depth of the ultrasound, and then apply coupling gel to the surface of cushion and probe. Then follow the operations below:

Press or button, make the probe contact closely with the patient. Be careful not to compress against the probe too tight so as to avoid any possible damage.

To search for focus in the ultrasound images, the figure obtained through watching

instructions, will be the actual distance from probe surface to focus (such as 90 shown in the image);

Activating the measuring function of the ultrasound (specific operation depending on ultrasound type), measure the distance between ultrasound image center line and probe surface, define the location of distance display, make "+" cursor of ultrasound image remain in this location, so this location will be the position of focus in the ultrasound image.

Referring to image direction in the ultrasound picture, press buttons of table motion on request, move stone location into focus in the ultrasound image; the process can be mastered soon through careful observation and repeated operation. During the process of localization, to reduce the height between human body and focus, first retract probe, then lower cushion to change the height.

WARNING

Protection against X ray or other safety issues should be noted when X ray machine is in use. For specific details please refer to safety report of X ray machine attached to the device.

During movement of table, the operator must notice the relative position between table and therapy head so as to avoid collision. As shown in figure 15: shockwave head A may collide with B, C and D surface of table, which will result in damage of the device and body injury.

CAUTION

In order to keep shockwave source from having over-pressure to the patient when shockwave source is in the top position for treatment, a pressure sensor will be installed inside shockwave source. Shockwave will stop move down as the pressure is above 100N.

CAUTION

The patient should get away from the device after water inside cushion is drained manually, in case the device get damaged or power failure.

During lithotripsy treatment

Press automatic triggering button to begin for shockwave release

WARNING

Before performing ESWL, the physician shall inform the patient about the treatment process. In particular, the patient must be informed about the noise caused by the release of shockwave. This is to prevent the patient from being startled and making involuntary movement during treatment.

Set triggering frequency of shockwave

Press adjustment button of triggering frequency to increase or reduce triggering frequency of shockwave. Generally speaking, a low frequency should be set at the beginning of treatment, then increase frequency gradually if the patient adapts to the shockwave well.

Set intensity of shockwave

Press energy up-down button to increase or reduce intensity of shockwave. Generally speaking, a low intensity (less than 12kV for electromagnetic shockwave source) should be set at the beginning of treatment, then increase the intensity gradually if the patient adapts to shockwave well.

CAUTION

Cushion should be close to the patient. After localization, the patient should not change his body position frequently. The relevant location between stones and focus must be checked frequently (usually using ultrasound localization for continuous monitoring after hundreds of shocks). If stone deviates from focus, stop shocks and then continue after accurate localization.

Under operation, three-dimension movement of table can not be operated in the state of triggering, it can be operated only when triggering stops. Shockwave cannot be released when 3D movement of table is operated, it can release only when the operation of 3D movement of table stop.

WARNING

The attending doctors must always monitor the patient throughout the treatment. If the patient felt uncomfortable or emergency happened, the treatment must be terminated immediately.

WARNING

The water temperature inside cushion may rise during the process of ESWL treatment. The attending doctors can press water cycling button on the operating panel for heat-exchange between water inside cushion and radiator, so as to cool it down.

WARNING

With electromagnetic shockwave source, do not set energy level above 12 at the early stage of each treatment. Triggering energy should be less than 5 with electro-hydraulic shockwave source at the beginning of the treatment.

CAUTION

If there is an emergency situation threatening safety, press emergency stop button directly to shut down the machine.

Conclusion lithotripsy treatment

1. Press stop button to terminate shockwave triggering
2. Press energy-down button repeatedly to drop the energy to zero
3. Press manual triggering button once or twice to release the remaining voltage
4. Press stop button to exit the starting state
5. Press power button to top position to cut off power
6. Clean the treatment part of the patient's skin
7. Clean the cushion

WARNING

Risk of electrical shock!
Do not skip the above-mentioned step 3 when power off the machine.

Judging stone fragmentation

After ESWL treatment, it is judged that the stone has been fragmented if the following phenomenon is observed:

a) Stone image becomes larger and lighter

b) Outline for stone changing significantly

c) Distribution for common sound-image in the monitor changing significantly

Gallstone fragmentation in the ultrasound image, apart from outline, there is no black bile between fragments and gallbladder wall.

Post-treatment processing

Attending physicians should perform anti-inflammation and stone-discharge after lithotripsy treatment.

a) Common complications and management

Common complications for ESWL including hematuria and skin erythema; in the event, it should be dealt with accordingly.

Hematuria: usually no special methods of treatment, it will be cured two or three days later. In individual cases proper amounts of anti-inflammation drug may be taken by mouth.

Skin erythema: it may be brought in the place of contact between the patient's body and cushion surface. Skin red, pain slightly caused by incomplete coupling between skin and cushion. No need for special treatment. In individual cases it can be handled with anti-inflammation drugs of external use.

In addition, quite a few patients will have stone street, which will bring renal colic. If this happens, stop treatment and ask urologists handle it according to conventional methods.

Equipment transportation

WARNING	**Overturning dangerous!** **Do not move HK.ESWL-V through more than 15 degrees slope or more than 2cm altitude difference.**
CAUTION	**Disconnect HK.ESWL-V from cables or water pipes before transportation.**

TECHNICAL SPECIFICATIONS

Introduction

The specification section provides technical information on:

- Dimensions and weight
- Power conditions
- Power capacity
- Environmental condition
- Shockwave performance parameter

All the specifications are standard values unless the tolerance is specified.

Dimension and weight

- Weight 1700kg
- Covered area(length*width) 5mx5m
- Height 2.5m

Power conditions

Power requirements AC 220V±10%、50±1Hz, internal resistance ≤0.3Ω;

The device requires the installation of protective grounding with separate grounding wire, grounding resistance ≤4Ω.

Power supply for the device is equipped with circuit breaker which can cut off phase wire and conform to requirements of IEC328, without self-resetting. Fuse capacity of the circuit breaker is ≥10A.

The device cannot be used in the same circuit in parallel with other large load equipment instantaneously (such as welding machine, elevator etc).

Power capacity

Power capacity	3kVA

WARNING **Do not use power supply for HK.ESWL-V which is not in accordance with the above-mentioned specifications.**

Environmental conditions

Indoor temperature	
During lithotripsy treatment	10～40℃
In storage (without water)	-40～70℃
In storage (with water)	1～70℃
Relative humidity	
During lithotripsy treatment	30～75% (no coagulation)
In storage	10-100%(including coagulation)
Atmospheric pressure	
During lithotripsy treatment	700～106hPa
In storage	500～1060hPa

CAUTION Water drained completely out of water tank in storage

Shockwave performance parameter

a) Principle	Electromagnetic
b)Shockwave contraction peak value	32. 5~50MPa
c) Shockwave expansion peak value	＜9MPa
d) Shockwave single pulse energy	53~128J（EM）
e) Shockwave pressure parameters	Pulse front ≤0.5μS； Pulse width ≤1μS； Focusing range: Radia ±7.5mm(Electromagnetic） Radial ±7.0mm(Electro hydraulic) Axial ±40mm (standard module tested by transducer is ±15mm)
f) Distance from second focus to shockwave source	130mm, standard (Users can choose one or more from 110mm、130mm、150mm)
g) Every energy level corresponding to kV value	Level 0 0kV, level 1 1kV, Level 2 2kV, level 3 3kV, Level 4 4kV, level 5 5kV, Level 6 6kV, level 7 7kV， Level 8 8kV, level 9 9kV, Level 10 10kV, level 11 11kV, Level 12 12kV, level 13 13kV, Level 14 14kV, level 15 15kV, Level 16 16kV

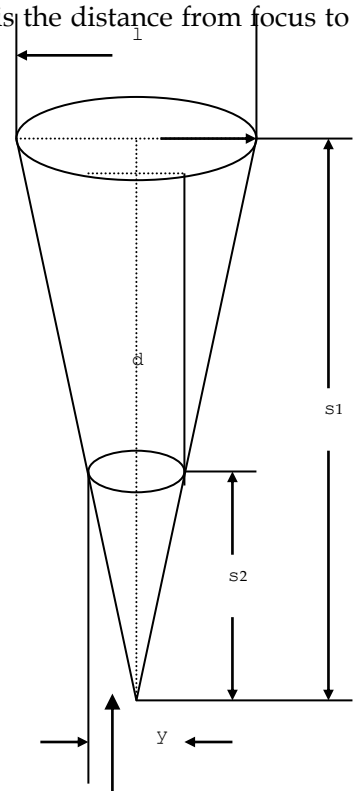
X ray performance parameter

1. X ray tube working voltage 50 kV～110 kV
2. X ray image resolution≥ 14 LP/mm（input screen size of image intensifier: 150 mm）
3. Additional filtration 0.9mmAL
4. Total filtration 2.9mmAL
5. X ray beam limiting assembly, that is collimator, fixed installation on top of tube window, not adjustable.
6. Distance from focus to image reception area 1100mm
7. Distance from focus to skin ≥ 300mm
8. X ray tube components, all components are not removable.
   a) Generator nominal frequency: 60Hz；
   b) Maximum tube voltage: 110kV; maximum tube current: 5mA；
   c) Non-removable material filtration: 2.0mmAL。
9. Tube and image intensifier are permanently fixed installation, reference axis is nonadjustable. Reference axis goes through the center of a circle of image reception area and is perpendicular to it.
10. Corresponding relations between X ray field and image reception surface
    Ray field the same plane as image reception surface is slightly larger than image reception area, and completely cover image reception area. The specific calculation sees below page
11. Tube overheat protection temperature 57℃
12. Tube maximum continuous heat-dissipation 60w

Relations between ray field and image reception area

1. Calculation method for ray field

   Fixed installation of calimator on top of tube window, and not adjustable; Axis extension line through tube focus and the centre of a circle of image reception area, and vertical to it; Figure 1, s1 is the distance from focus to image reception surface, s2 is the distance from focus to the top of calimator. "1" is the diameter of ray field,
   "d" is the inner diameter of calimator,
   then 1=d* s1/s2, s1=1100mm, s2=100mm, d=14mm,
   Calculation comes as 1=154mm



2. Relations between ray field and image reception surface

   Diameter of image reception surface is a, a=230mm,
   as shown in figure 2, outer circle is ray field,
   inner circle is image reception surface

   $x' = l - a = 4mm \leq s1 \times 3\% = 33mm$

   $y' = l - a = 4mm \leq s1 \times 3\% = 33mm$

   $x' + y' = 8mm \leq s1 \times 4\% = 44mm$

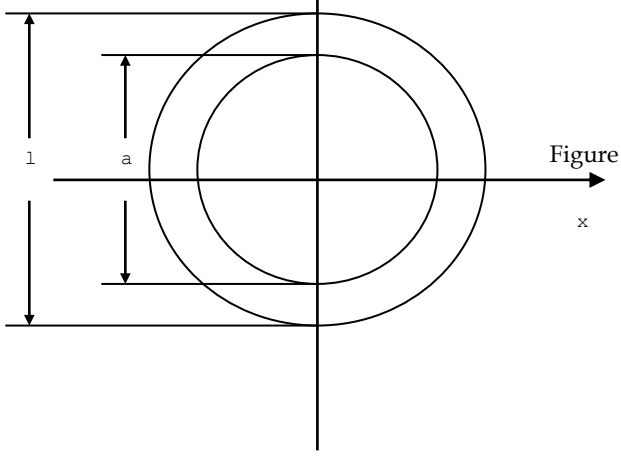


Figure 1

Figure 2

Localization system parameter

X ray localization (C arm single X ray)

X ray images resolution:

Input screen size of image intensifier	Resolution
150 mm	≥ 14LP/㎝

CAUTION

Anyone who needs to enter the working main room of fluoroscopy examination must wear protective clothing against X ray.

Protective shield with size no less than 60cm*60cm*200cm must be placed when close to X ray tube within 2m.

CAUTION **In order to avoid patient absorb too much dose of X ray, the patient should be put as far as possible away from the focus so long as the image is clear enough for treatment.**

Ultrasound localization

- Radial error between probe extension and focus ≤±2mm;
- Distance-measuring error between probe surface and focus ≤±2mm;
- Ultrasound images resolving power vertical ≤2mm; horizontal ≤3mm;
- Detection penetration ≥18cm;

Mechanical motion distance

Table

- Up/down 120mm
- Backward/forward 100mm
- Right/left 100mm

C arm

- Rotation motion Clockwise 30°, counterclockwise 20°

Membrane arm

- Rotation motion 180°

Shockwave source

- Orbital motion 35°

Localization accuracy

For HK.ESWL-V, deviation between actual focus and localization mark of monitor is ≤3 mm

Localization system

Ultrasound localization (with configuration of CTS-360A ultrasound)

- Radial error between probe axis extension and focus is ≤±2mm;
- Distance-measuring error between probe surface and focus ≤±2mm;
- Ultrasound images resolution vertical ≤2mm; horizontal ≤3mm;
- Detection penetration ≥10cm;
- Probe specification 3.5MHz convex or linear

MAINTENANCE

Introduction

The maintenance section informs you about:

- Safety inspection
- Stability test
- Focus calibration
- Adding water to water tank
- Maintenance
- Replacement of fuse
- Cleaning and disinfecting
- Disposal

Safety inspection

Periodic safety check

Safety check means maintenance and adjustment to HK.ESWL-V in regular time.
Safety check can ensure safe and reliable use of HK.ESWL-V.
Shenzhen Huikang Medical Apparatus Co., Ltd recommends safety inspection to HK.ESWL-V every year.

Temporary safety check

Safety check must be conducted if one of the followings occurs:

- Installation
- Re-installation
- Maintenance
- Replacement of consumables

Shenzhen Huikang Medical Apparatus Co., Ltd can provide safety check list free of charge upon request.

WARNING

Only authorized personnel may perform safety inspection. Authorized personnel are the persons who have been trained by Shenzhen Huikang Medical Apparatus Co., Ltd or a company authorized by Shenzhen Huikang Medical Apparatus Co., Ltd. Safety check was conducted by un-authorized personnel may result in injury to persons or damage to the device.

Stability test

Apart from consumables, service life designed for this device is eight years. During the period of service life, to avoid danger caused by abnormal aging, stability test should be conducted in regular time in the following ways:

a) Simulation lithotripsy

Electromagnetic shockwave source: put simulation stone with the size of round 1cm in the focus of shockwave, set energy 15, discharge 300 times, the maximum stone fragmentation within 3mm

b) Noise detection

It should not exceed 90 decibels with noise detector when the machine is running (discharge noise not included)

Focus calibration

1、Ultrasound localization calibration

① Check regularly whether indicated value for extension & contraction limit distance of probe changes or not.; If changes are more than 3mm, it should be re-adjusted. For example, when delivery from factory, value for "probe distance-measuring display" is set as below:

Probe extension limit: 25 mm (upper limit)

Probe contraction limit: 125 mm (lower limit)

② check method:

i. Localization pin installed in the shockwave source, localization pin inserted inside ultrasound assembly (inside ultrasound probe holder)

ii. Make top end of motion frame of ultrasound up-down assembly in the place of "2.5cm"

iii. Check whether the two pins contacts tightly; it needs to be adjusted if not.

③ adjustment method as follows:

Extending probe to the top limit, adjust four hand-driving screws under ultrasound localization assembly, adjust direction of ultrasound assembly till ultrasound localization pin close to localization pin of shockwave source tightly, tightening the four screws after adjustment.

2、X ray localization calibration

① check method:

a) Localization pin installed in the wave source

b) Press button, make C arm in the place of orthographic projection, exposure and mark the pinpoint in the monitor screen

c) Press button, exposure and observe in the monitor screen if pinpoint

and mark overlap; if overlap, it means focus mark is correct; if not, re-adjustment and calibration needed. (only by authorized personnel)

CAUTION To check accuracy of ultrasound localization at least once a month; if reduced limit distance changed from indicator value exceed 1mm during ultrasound localization, re-calibration should be conducted.

Adding water to water tank

Water must be added into water tank before initial use. Open infusion cover and add water to the tank till it is full.
Usually, add water to the tank by every 2 or 3 months. Users should open the infusion cover regularly to check if water is needed.

Replacement of consumables

There are 5 kinds of consumable for HK.ESWL-V as follows:

A. Capacitor
B. Electromagnetic disk
C. Acoustic lens
D. Electrode (for EH shockwave source only)
E. Cushion

(for electromagnetic shockwave source, consumables are A, B, C, E;
for electro hydraulic shockwave source, consumables are A, D, E)

Only authorized personnel may replace item A, B or C.
Users can replace the cushion in the following ways:

1) Discharge all the water in the cushion.
2) Cut off power of the machine.
3) Loose the nuts on the metal loop that binds the cushion with a screw-driver.
4) Remove the metal loop and cushion.
5) Replace with new cushion and fix the metal loop again. Tighten the nuts with screw driver.

Users can replace the electrode in the following ways:

1) Discharge all the water inside cushion;
2) Cut off power of the machine;
3) Loose nuts on the metal loop that binds cushion with a screw driver;
4) Remove cushion, then electrodes with special tools;
5) Replace with new electrodes and tighten them to prevent leakage;
6) Install cushion and fix the metal loop;
7) Tighten the nuts with screw driver.

WARNING

Huikang is the only legal supplier of consumables. The loss due to abuse of consumables not provided by Huikang will be borne by users. Replacement of Electromagnetic disk can be conducted by the personnel of Huikang or the personnel authorized by Huikang.

Maintenance

Maintenance for HK.ESWL-V should be performed by authorized personnel only.

WARNING

Only authorized personnel may perform maintenance for HK.ESWL-V. Authorized personnel are the persons who have been trained by Shenzhen Huikang Medical Apparatus Co., Ltd or a company authorized by Shenzhen Huikang Medical Apparatus Co., Ltd. Safety check conducted by un-authorized personnel may result in injury to persons or damage to the device.

When powering off the machine for high voltage maintenance, capacitor must be discharged first to avoid electric shock.

Maintenance operation can be carried out only as long as there are at least two persons at scene.

Common faults judgment and disposal

Faults	Analysis of cause	Processing method
Too much misfires	Usually caused by the change in capacitor performance. There will also be misfire if capacitor comes to the life time.	Replacement of capacitor
Much succeeding shocks, uncontrolled discharge continuously	Usually caused by smaller triggering discharge gap, volts-increasing too high or capacitor to life	Replacement of capacitor
Triggering without discharging	Blue flash seen from cushion (Electromagnetic)	Electromagnetic disk has breakdown

Replacement of fuse

Fuse number and corresponding virtual value.

Power off the machine, follow up requirements of the table or schematic diagram to replace the fuse of same specification conforming to relevant regulations.

Cleaning and disinfecting

1. Cut off the power supply.
2. Clean and disinfect the device in the following ways:
   - Cleaning: wipe off dirt and blotches on the accessible surface of the device with a sponge or soft cloth.
   - Disinfecting: After cleaning, wipe the outer surface of the device with sponge or soft cloth dipped in disinfectant (Do not dip too much disinfectant liquid in the sponge or soft cloth to avoid splashing drops cause device failure or danger). Available disinfectants including: medical alcohol, formalin and other disinfectants. Before each treatment, the following application parts of the device should be disinfected on surface:
     1) Therapy head (including cushion)
     2) Operating panel
     3) Any other parts that may contact with patients or operators during the treatment.

WARNING

Do disconnect the power supply before cleaning and disinfection.
Do not use the gas or spray disinfectants for cleaning and disinfection.
Do not allow liquid penetration inside the device, especially into electrical parts.
Do follow safety instructions given by manufacturer of disinfectants as well as detergent.
Do follow legal regulations concerning disinfection and cleaning.

Disposal

At the end of service life, HK.ESWL-V must be disposed of in accordance with local laws and regulations.

The different materials, such as packing, electronic scrap, plastics, metal, wastewater etc, must be disposed of in accordance with relevant national laws and regulations.

For inquiries regarding the avoidance of personal injury and/or

contamination to the environment, please contact Shenzhen Huikang Medical Apparatus Co., Ltd before HK.ESWL-V is out of its term in use.

Daily maintenance

a) Maintain lubrication of mechanical parts such as track, bearing etc, wipe off dust and cleaning

b) Check whether each grounding wire is connected tightly at least one time a month

c) Check whether each ground wire is connected normally and tightly before starting each time, whether there is abnormal sound after starting, whether there is abnormal phenomenon during trial running

d) Keep dry against moisture

Inspection methods of focus location and focus size

1、Test for pulse width, pulse front and focus range of shockwave

1、1 Shockwave performance index of HK.ESWL-V

1.1.1 Focal pressure 20 MPa～50 MPa

1.1.2 Pulse width ≤1 μs

1.1.3 Pulse front ≤0.5 μs

1.1.4 Focus range

a) Radial ±7.5 mm

b) Axial ±40 mm

1、2 Test instruments

a) Pressure transducer with frequency response not less than 2MHz, pressure-measuring not less than 100MPa.

b) Digital storage oscillograph with storage sampling frequency not less than 20MHz or memory oscillograph with frequency response not less than 50MHz

1、3 Test procedure

a) Test ten consecutive times when putting transducer in the focus position, there should be at least 40% shockwave shape which comply with “1.1.1, 1.1.2, 1.1.3” requirements.

b) Double transducers test or multi-transducers test, focus range of shockwave should conform to “1.1.4” requirements.

2、 Focus distance test

2.1 Focus distance

a) The distance from focus to port plane of corresponding lens is respectively 130 mm±2 mm

2.2 Test method

Measuring the distance between simulation focus and reflector plane with square ruler under "2.1" requirements

CAUTION **Please contact us timely in case any non-conformance is found in pulse width, pulse front, focus range and focal distance of focus shockwave.**

Inspection procedure for why focus point caused by error motion of table and inaccuracy of ultrasound position device does not coincide with stone location

3.1 Ultrasound localization calibration

① Regularly checking if there will be change to the figure indicated by limit distance between probe extent and shrink. If the changes are more than 3mm, then re-adjustment should be done. For instance, the setting figure of probe distance-measuring display comes as follows as the machine leave factory:

Probe extent limit:____25_____mm（Upper limit）

Probe shrink limit:___125____mm（Lower limit）

② Inspection:

a. Making localization pin be installed in the shockwave source, then ultrasound localization pin in the ultrasound device (inside probe frame)

b. The top end of motion frame of ultrasound up-down device put in the 2.5cm position of calibrated scale;

c. For the time being checking if these two points of pin stay in tight closely, if not, they need to be adjusted.

③ Adjustment method as follows:

Probe extended to the upper limit, unscrew the four bolts of ultrasound localization device base, adjust direction of ultrasound localization device, lastly, make the two pin points stay in tight closely, then tighten the screws after adjustment.

3.2 Inspection procedure for observing in the ultrasound display why focus point caused by error motion of table do not coincide with stone location

3.2.1 To understand location of stones, number, shape with ultrasound probe before patients on the table, and to set detection direction, location of ultrasound probe initially, marked on the human body if necessary.

3.2.2 Check regularly if the calculi shown in the ultrasound monitor coincide with "+" mark of original stone or not?

CAUTION **If physical variables deviation which will bring patients increasing risk are found, please contact us timely.**

Test procedures for physical variables deviation bring increasing risk of patients

1. Inspection procedures for actual energy of shockwave higher than displayed value.
   (1) Test directly high volts output of shockwave with 30kV high volts test meter;
   (2) Test primary high-voltage output of high-volts transformer with the 300V AC ammeter (see circuit schematic diagram HK-V-SCH-005, TU output), 100 times of measurement value shall be high-voltage output value of shockwave. For energy one high-voltage output of shockwave is 10kV±5%; for energy 10, high-voltage output is 16kV±5%, which means there is no deviation.

2. The actual fluoroscopy mA of X ray not in accordance with displayed value.
   Remove CP3 jumper wire of S19 board (see circuit schematic diagram HK-V-SCH-006, S19-CP3), connected to the mA ammeter, observe reading of mA ammeter during fluoroscopy, deviation within 10% between reading and displayed value shall be taken as in accordance.

CAUTION

Please contact us in time for solutions if physical deviation beyond the above-mentioned value are found.

REFERENCES

Introduction

The references section contains:

- List of abbreviation
- List of glossary

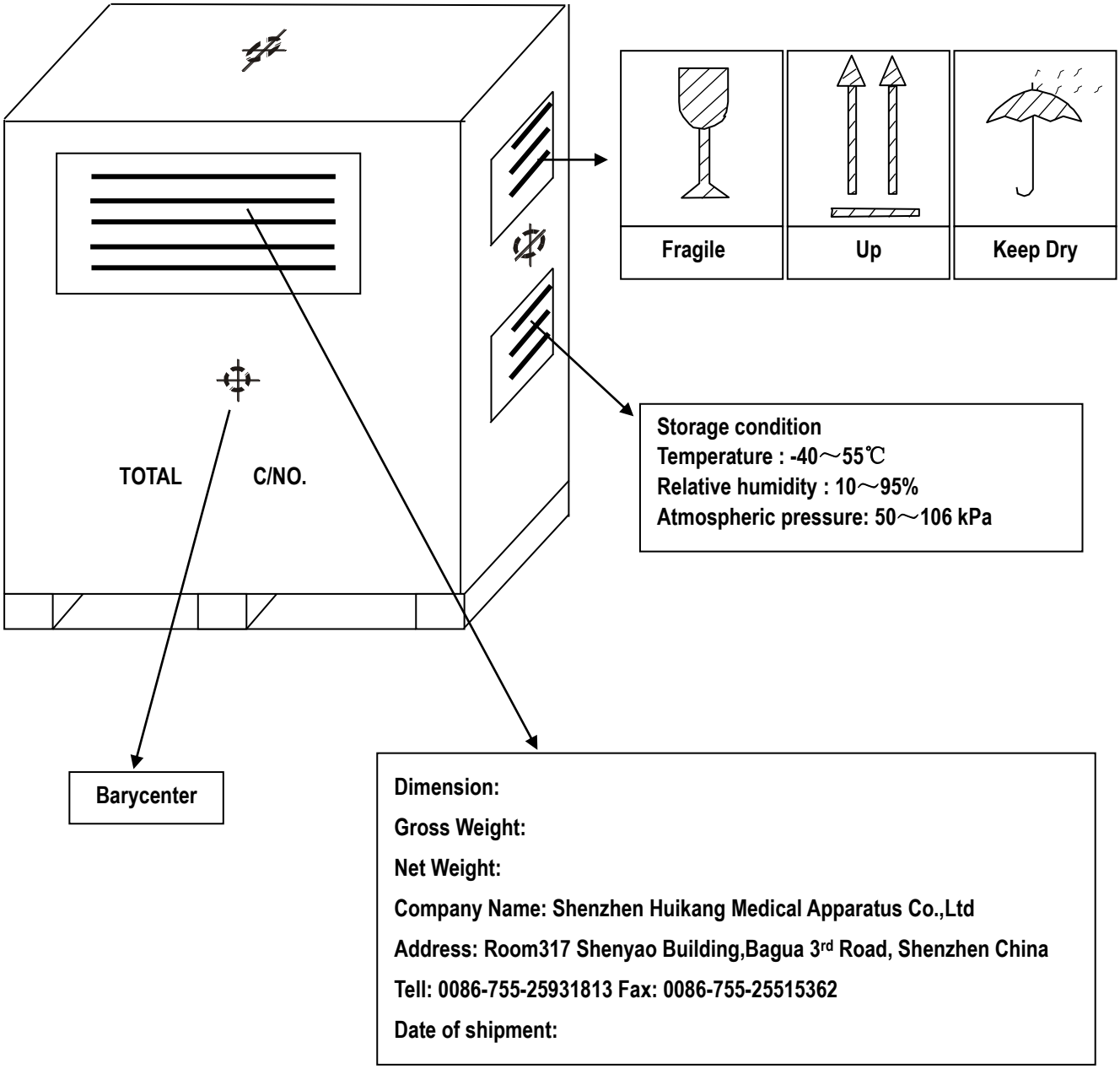
Abbreviation

Abbreviations	Meaning
ESWL	HK.ESWL-V (Extracorporeal Shockwave Lithotripter)

Glossary

Glossary	Meaning
Shock wave pan	The pan-shaped shockwave generator.
Coupling gel	Medium used to eliminate air between patients' skin and coupling cushion. Air impairs effects of shockwave.
Electrode	Electro-hydraulic shockwave generator.
Cushion	Silicon gel pouch used to pass shockwave between patients and shockwave source after filling water.
Focus	Through focusing by the lens, shockwave reaches the maximum energy.

Transport Sign



For further details of the Extracorporeal Shockwave Lithotripter (HK.ESWL-V), please contact us freely.